DAVIDE BALDELLI

PhD researcher at Polytechnique Montréal and Mila

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EDUCATION

MILA and Polytechnique Montréal *Jan 2025 - Present*
PhD in Computer Engineering (GPA: 4.0/4.0)
- Supervisors: Sarath Chandar, Amal Zouaq.

University of Bologna *Sep 2021 - Oct 2023*
MSc in Artificial Intelligence (Grade: 110/110 cum laude)
- Thesis: A TWO-step LLM Augmented distillation method for passage Reranking. [thesis]

University of Florence *Sep 2017 - Mar 2021*
BSc in Mathematics (Grade: 110/110)
- Thesis: Big Deviations for Markov Chains. [thesis]

PUBLICATIONS

Cross-Model In-Context Neurofeedback Reveals (Limited) Activation Control in LLMs *Under review*
Davide Baldelli, Robyn Kozierok, Mirko Bronzi, Damiano Fornasiere [paper] [code]

ϕTD: Distributional Reinforcement Learning using Characteristic Functions *NeurIPS 2026*
Thomas Mousseau, Tyler Kastner, **Davide Baldelli**, Amir-massoud Farahmand [paper]

Probabilistic Calibration Is a Trainable Capability in Language Models *Under review*
Davide Baldelli, Sruthi Kuriakose, Maryam Hashemzadeh, Amal Zouaq, Sarath Chandar [paper] [code]

LLMs Can't Play Hangman: On the Necessity of a Private Working Memory for Language Agents *CoLLAs 2026 (Oral)* *MemAgents Workshop @ ICLR 2026*
Davide Baldelli, Ali Parviz, Amal Zouaq, Sarath Chandar [paper] [code]

CADmium: Fine-Tuning Code Language Models for Text-Driven Sequential CAD Design *TMLR 2026*
Prashant Govindarajan*, **Davide Baldelli***, Jay Pathak, Quentin Fournier, Sarath Chandar [paper] [code]

TWOLAR: A TWO-Step LLM-Augmented Distillation Method for Passage Reranking *ECIR 2024 (Oral)*
Davide Baldelli, Junfeng Jiang, Akiko Aizawa, Paolo Torroni [paper] [code]

*Equal contribution.

EXPERIENCE

MATS (ML Alignment & Theory Scholars) *Jun 2026 - Present*
Research Fellow (10.0 + 10.1 Extension)
- Mentored by Mirko Bronzi and Damiano Fornasiere (LawZero).
- Selected for the Summer 2026 cohort (~5% acceptance rate); continued into the extension phase from Sep 2026.
- Studying introspection in language models; first output: the Cross-Model In-Context Neurofeedback preprint.

ARENA 7.0 (Alignment Research Engineer Accelerator) *Jan - Feb 2026*
Participant
- Technical AI safety program focused on building ML engineering skills for alignment research.

Loop AI Labs Inc. *Sep 2022 - Dec 2024*
Machine Learning Engineer
- Built time-series forecasting models (e.g., power consumption) and email classification systems for client projects; part-time, then full-time from Sep 2023.

- Co-supervised a master's thesis titled "Self Assessment Tool for Medical Exam Candidates" in collaboration with the University of Padova.

National Institute of Informatics (NII) *Mar 2023 - Aug 2023*
Research Intern, NII International Internship Program
- Selected through the University of Bologna's nomination; supervised by Akiko Aizawa and Paolo Torroni.
- Developed TWOLAR (ECIR 2024, oral), a 3B LLM-distilled passage reranker that set the state of the art on BEIR at publication.

SERVICE AND LEADERSHIP

INF8245AE Machine Learning – Polytechnique Montréal *Fall 2026*
Teaching Assistant

Fatima Al-Fihri Predoctoral Fellowship *Fall 2026*
Research Mentor
- Mentoring a fellow on a machine learning research project through the Fatima Institute for Global AI Research, a program supporting prospective PhD applicants from low- and middle-income countries.

SPAR (Supervised Program for Alignment Research) *Fall 2026*
Research Mentor
- Mentoring two research fellows on building robust activation monitors for language models.

CoLLAs 2026 (Conference on Lifelong Learning Agents) *Nov 2025 - Present*
Web Communication Chair
- Managing the official conference website and digital presence for the 2026 edition.

MILA *Summer 2025*
Teaching Assistant - Summer High School Internships
- Designed and delivered educational workshops on Machine Learning and Probability for high school students.

UILO – Career Orientation Platform *Sep 2024 - Present*
Co-Creator and Technical Lead
- Built UILO, a career orientation platform for post-secondary education with university and career databases and psychometry-based recommendations.

Belle Parole Arci APS *Nov 2023 - May 2026*
Co-Founder, Non-Profit Association
- Organized community events and communications, built the organization's website **belleparoleaps.it**, and secured grant funding for organizational activities.

ADDITIONAL INFORMATION

Reviewing	ICLR 2027; CoLLAs 2026; MemAgents Workshop at ICLR 2026.
Talks	CADmium, Science Coffee at the ESA Advanced Concepts Team (Feb 2026).
Recognition	Selected Participant, Y Combinator AI Startup School (Jun 2025).
Languages	English (professional proficiency); Italian (native proficiency).